

Assignment-2
Graphic Equaliser

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Music and Audio Programming – ECS7012P

Introduction

The assignment focuses on the implementation and analysis of graphic equalizer using a JUCE-based audio application. A graphic equalizer allows control on amplitude of different frequency bands within an audio signal, enabling targeted boosting or reducing of low, mid, and high frequency components. These features make it a useful tool for adjusting the tonal balance of audio.

A white noise was used as the input to verify the system because of its uniform distribution across the frequency spectrum, which makes it suitable for observing frequency-based changes. The resulting output was analyzed using spectral plots after applying gains to specific frequency bands. The changes in the frequency content confirms that the equalizer modifies the signal as intended.

Implementation

The graphic equalizer was implemented as a JUCE-based audio application with a user interface consisting of three rotatory sliders controlling the gain of low, mid, and high frequency bands. This interface also includes buttons for loading, playing, and stopping the audio playback, allowing the user to test the equalizer in real time. The slider values are linked to internal parameters using the `AudioProcessorValueThreeState`, ensuring that changes in the user interface are directly reflected in the audio processing stage.

The processing of the audio is carried out in the `processBlock()` function, where the input signal is either read from a loaded audio file or generated as a white noise if no file is loaded. The gain values from the sliders are retrieved in decibels and converted to linear scale before being applied to the filters. Three types of IIR filters are used to implement the equalizer: a low shelving filter for low frequencies, a peaking filter for mid frequencies, and a high shelving filter for high frequencies.

These filters are applied sequentially to each audio channel, effectively forming a series processing chain. Separate filter instances are created for each channel to ensure correct stereo processing. The gain applied to each filter determines how much the corresponding frequency band is amplified. By adjusting the slider values, the user can control the amount of boost applied to each frequency range, and the processed signal is then sent to the output in real time.

Testing and Analysis

The performance of the graphic equalizer was evaluated using white noise as the input signal. White noise was selected because it contains a uniform distribution of energy across all frequencies, making it suitable for analysing how different frequency bands are affected by gain adjustments. This allows any changes introduced by the equalizer to clearly see the frequency domain.

The equalizer was tested by applying gain to specific frequency bands using the user interface controls. In particular, low-frequency and high-frequency boosts were applied separately while keeping other bands unchanged. The output signals were recorded using Audacity and analysed using spectral plots. By comparing the spectra of the original signal and processed signals, the effect of the equalizer on different frequency regions could be clearly identified.

This analysis focuses on observing how the magnitude of the signal changes across the frequency spectrum after applying gain. Any increase in amplitude within a specific frequency range indicates that the corresponding filter is functioning correctly. This approach provides a clear and effective method to verify that the equalizer modifies the signal as intended.

Results

The frequency spectra of the original and processed noise signals were analysed to evaluate the performance of the graphic equalizer. As shown in fig.1, the spectrum of the original audio i.e white noise shows a relatively uniform distribution of energy across all frequencies, which is expected due to its flat spectral characteristics.

When a low-frequency boost of +10dB was applied, the resulting spectrum in fig.2 shows a clear increase in magnitude in the lower frequency region, while the higher frequency region remains largely unchanged. This demonstrates that the low shelving filter is effectively amplifying the low-frequency components of the signal.

Similarly, when a high-frequency boost of +10dB was applied, the spectrum in fig.3 shows a noticeable increase in magnitude in the higher frequency region, with minimum effect on the lower frequencies. This confirms that the high shelving filter is correctly enhancing the high-frequency components.

Overall, the comparison between fig1, fig2, and fig3 shows that the graphic equalizer successfully modifies the amplitude of specific frequency bands while leaving other regions relatively unaffected, confirming correct working of the system. The spectral plots for the original and processed signals are shown below.

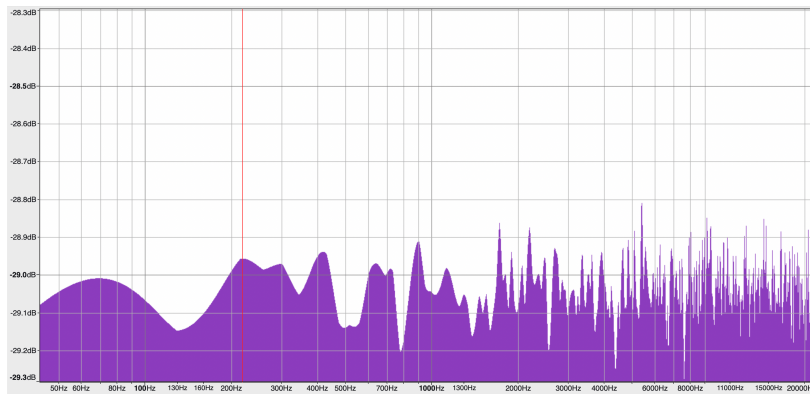


Figure1: Frequency spectrum of the original white noise signal showing a relatively flat distribution of energy across all frequencies.

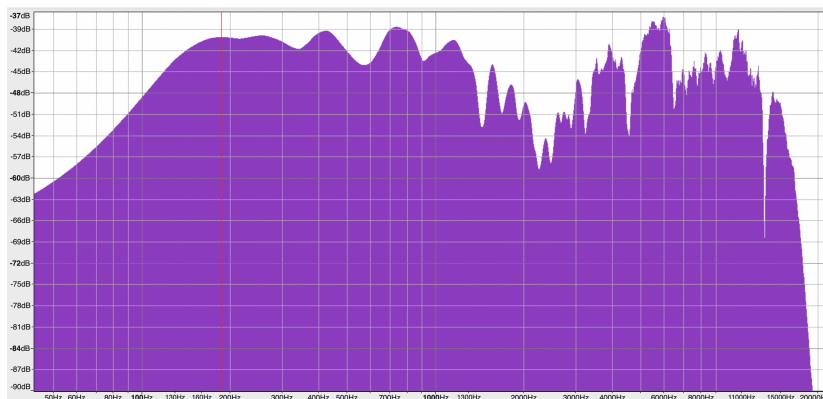


Figure2: Frequency spectrum after applying a +10dB low-frequency boost, showing increased energy in the lower frequency region.

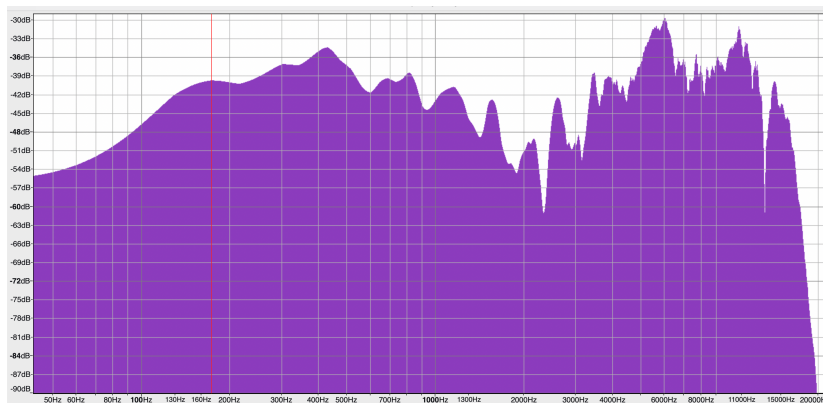


Figure3: Frequency spectrum after applying a +10dB high-frequency boost, showing increased energy in the higher frequency region.

Conclusion

The graphic equalizer plugin was successfully developed and tested using white noise as the input signal. The result demonstrates that adjusting the low and high gain parameters effectively modifies the corresponding frequency regions of the signal. The spectral analysis confirmed that boosting low frequencies increases energy in the lower range, while boosting high frequencies enhances the higher frequency components.

Overall, the plugin performed as expected, with clear and noticeable changes in both audio perception and frequency spectra. This validates the correct implementation of the equalizer and its ability to control different parts of the audio spectrum.